



Sliced Optimal Transport

Kimia Nadjahi
kimia.nadjahi@ens.fr

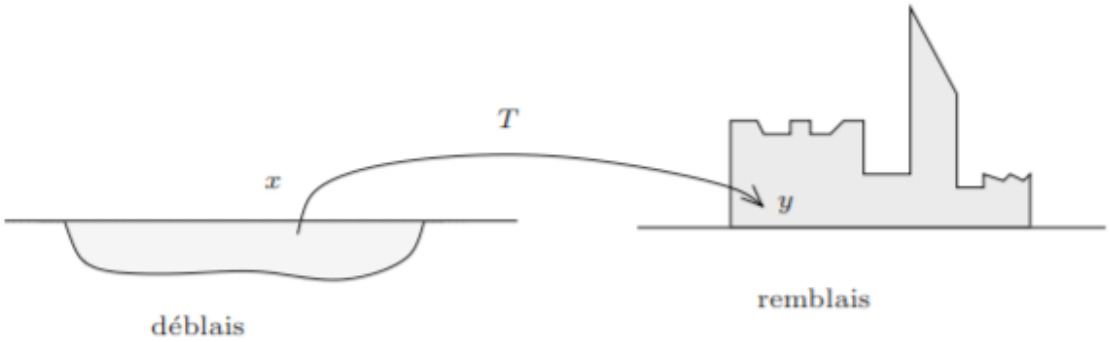
Harvard University
CS 2840: Computational Optimal Transport for Machine Learning

October 9, 2025

Reminders on Optimal Transport

The First OT Problem (Monge, 1781)

How do you move sand into a pit at minimal cost?

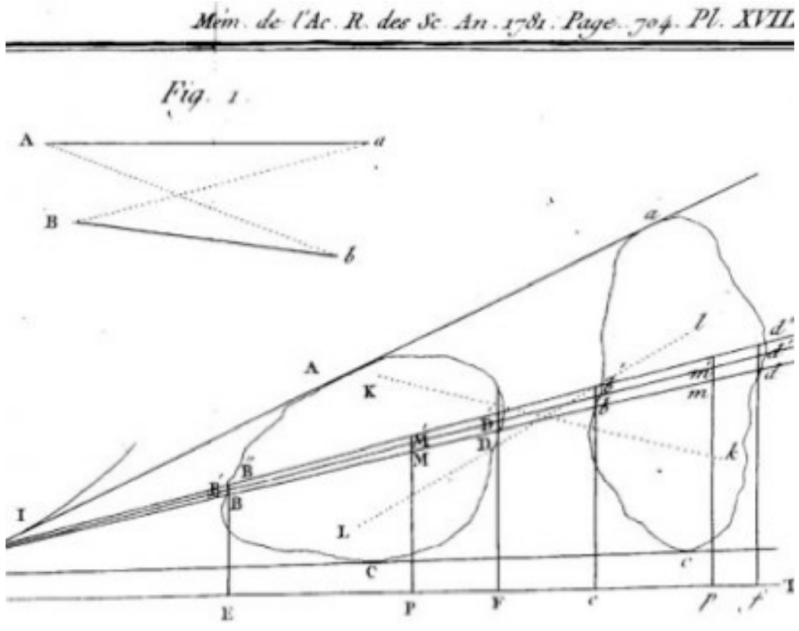


MÉMOIRE
SUR LA
THÉORIE DES DÉBLAIS
ET DES REMBLAIS.

Par M. MONGE.

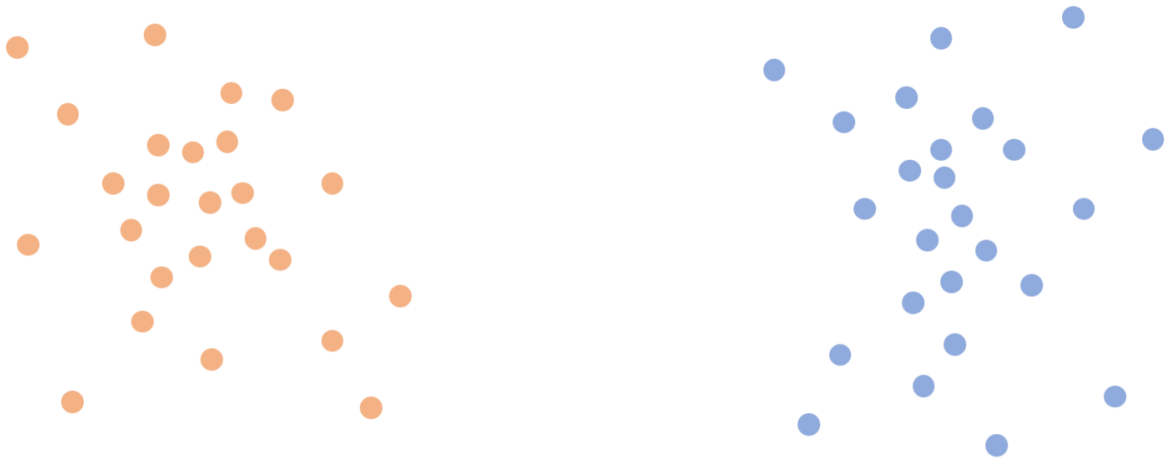
LORSQU'ON doit transporter des terres d'un lieu dans un autre, on a coutume de donner le nom de *Déblai* au volume des terres que l'on doit transporter, & le nom de *Remblai* à l'espace qu'elles doivent occuper après le transport.

Le prix du transport d'une molécule étant, toutes choses d'ailleurs égales, proportionnel à son poids & à l'espace qu'on lui fait parcourir, & par conséquent le prix du transport total devant être proportionnel à la somme des produits des molécules multipliées chacune par l'espace parcouru, il s'ensuit que le déblai & le remblai étant donnés de figure & de position, il n'est pas indifférent que telle molécule du déblai soit transportée dans tel ou tel autre endroit du remblai, mais qu'il y a une certaine distribution à faire des molécules du premier dans le second, d'après laquelle la somme de ces produits sera la moindre possible, & le prix du transport total sera un *minimum*.



Mémoire sur la théorie des déblais et des remblais, Monge, 1781.

A Versatile Tool for Comparing Distributions



Assignment between point clouds

The Wasserstein distance

$$\mathbf{W}_2(\mu,\nu) \triangleq \left(\inf_{\pi \in \Gamma(\mu,\nu)} \int_{\mathcal{X}\times\mathcal{Y}} \|x-y\|^2 \, \mathrm{d}\pi(x,y) \right)^{1/2}$$

with $\Gamma(\mu,\nu)$: set of distributions on $\mathcal{X}\times\mathcal{Y}$ with marginals μ and ν

A Versatile Tool for Comparing Distributions



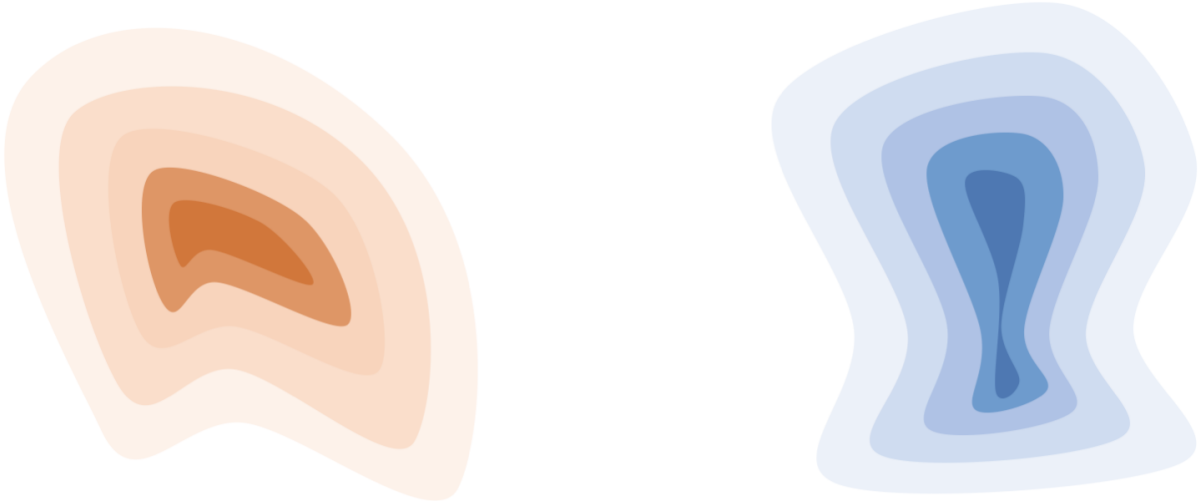
Transport between discrete distributions

The Wasserstein distance

$$\mathbf{W}_2(\mu,\nu) \triangleq \left(\inf_{\pi \in \Gamma(\mu,\nu)} \int_{\mathcal{X}\times\mathcal{Y}} \|x-y\|^2 \, \mathrm{d}\pi(x,y) \right)^{1/2}$$

with $\Gamma(\mu,\nu)$: set of distributions on $\mathcal{X}\times\mathcal{Y}$ with marginals μ and ν

A Versatile Tool for Comparing Distributions



Transport between continuous distributions

$$\rho_\mu(x),\ \rho_\nu(x)$$

The Wasserstein distance

$$\mathbf{W}_2(\mu,\nu) \triangleq \left(\inf_{\pi \in \Gamma(\mu,\nu)} \int_{\mathcal{X}\times\mathcal{Y}} \|x-y\|^2 \,\mathrm{d}\pi(x,y) \right)^{1/2}$$

with $\Gamma(\mu,\nu)$: set of distributions on $\mathcal{X}\times\mathcal{Y}$ with marginals μ and ν

A Versatile Tool for Comparing Distributions



Transport between continuous/discrete distributions

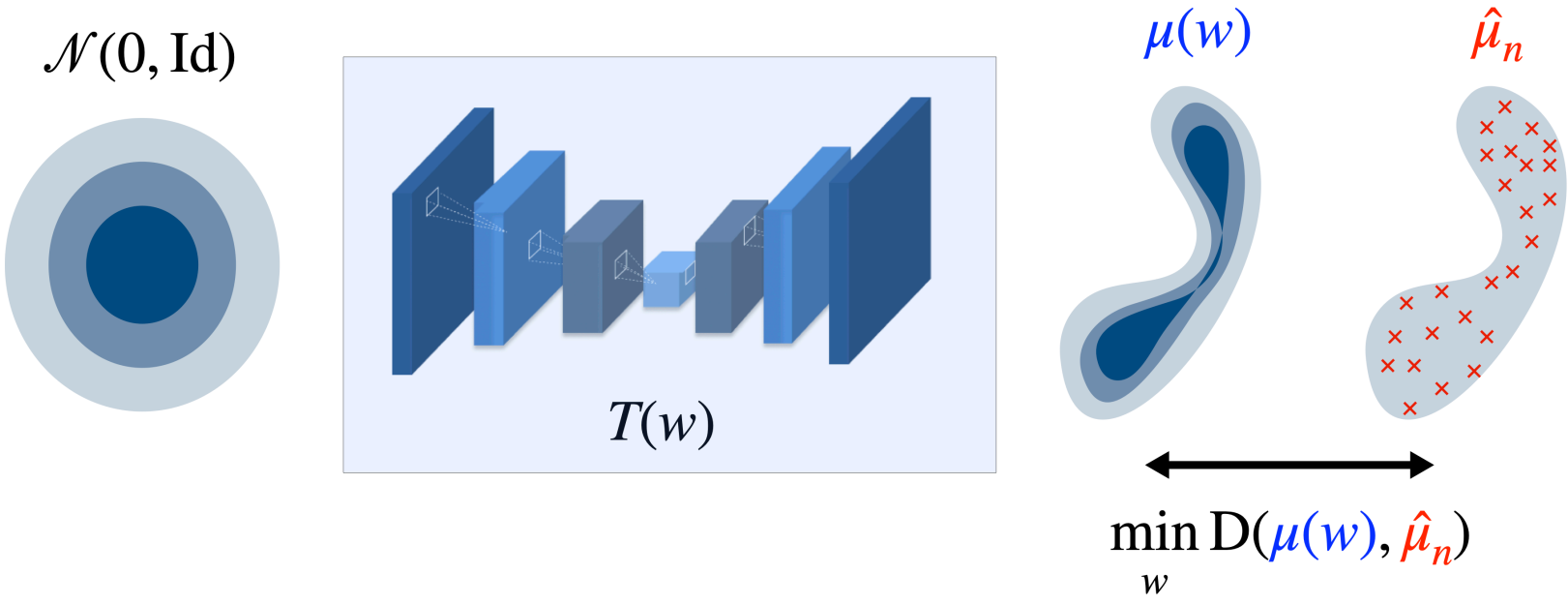
$$\rho_\mu(x),\ \nu = \sum_{j=1}^{m} b_j \delta_{y_j}$$

The Wasserstein distance

$$\mathbf{W}_2(\mu,\nu) \triangleq \left(\inf_{\pi \in \Gamma(\mu,\nu)} \int_{\mathcal{X}\times\mathcal{Y}} \|x-y\|^2 \, \mathrm{d}\pi(x,y) \right)^{1/2}$$

with $\Gamma(\mu,\nu)$: set of distributions on $\mathcal{X}\times\mathcal{Y}$ with marginals μ and ν

OT for Machine Learning



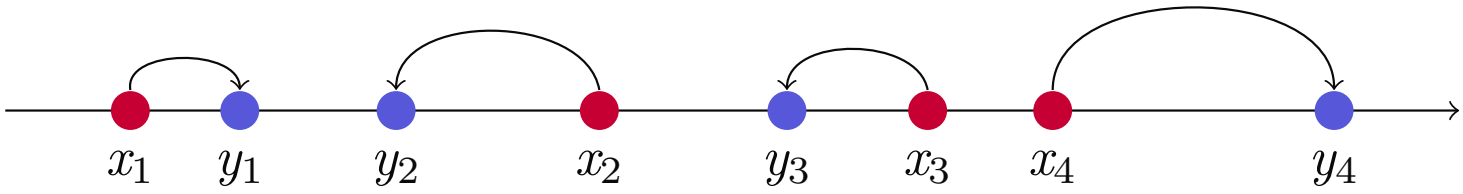
Generative models

Some OT-based deep generative models:

- Wasserstein GANs [Arjovsky et al., 2017]: GANs with objective function based on the dual of $\mathbf{W}_1(\mu(w), \hat{\mu}_n)$
- Wasserstein Auto-Encoders [Tolstikhin et al., 2018]: Auto-encoders trained by minimizing a regularized $\mathbf{W}_1(\mu(w), \hat{\mu}_n)$

OT in One Dimension: Sort and Match

Let $\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$ and $\nu = \frac{1}{n}\sum_{i=1}^{n} y_i$ with $x_i, y_i \in \mathbb{R}$



source (μ) target (ν)

$$\mathbf{W}_2^2(\mu, \nu) = \frac{1}{n}\sum_{i=1}^{n} |x_{(i)} - y_{(i)}|^2$$

with $x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(n)}$, and $y_{(1)} \leq y_{(2)} \leq \cdots \leq y_{(n)}$

Fast to compute: Just sort $\rightarrow \mathcal{O}(n\log(n))$

OT in Multi-Dimensions

Now, μ, ν are distributions on $\mathbb{R}^d$

Discrete measures

Let $\mu_n = \sum_{i=1}^n a_i \delta_{x_i}$ and $\nu_n = \sum_{j=1}^n b_j \delta_{y_j}$

$$\mathbf{W}_2^2(\mu_n, \nu_n) = \min_{\pi \in \Pi(a,b)} \left\{ \langle C, \pi \rangle \triangleq \sum_{i=1}^n \sum_{j=1}^n C_{ij}\, \pi_{ij} \right\}$$

with $C \in \mathbb{R}^{n \times n}$ the cost matrix s.t. $C_{ij} = c(x_i, y_j)$, and

$$\Pi(a, b) \triangleq \{\pi \in \mathbb{R}_+^{n \times n} \text{ s.t. } \pi \mathbb{1}_n = a,\ \pi^T \mathbb{1}_n = b\}$$

OT is a linear program

- Linear program with n^2 variables and $2n$ constraints
- Classical solvers: Hungarian algorithm, interior-point method
- Computational complexity $\mathcal{O}(n^3)$

Standard OT: Too Costly in ML

Computational bottleneck: Computing $\mathbf{W}_2(\mu_n, \nu_n)$ costs $\mathcal{O}(n^3)$
$\Rightarrow$ Expensive for modern ML pipelines

Example: Training OT-based neural networks

- Mini-batch $n = 256 \Rightarrow 1$ evaluation of $\mathbf{W}_2(\mu_n, \nu_n) \sim 10^7$ operations
- Needed at every training step, over thousands of iterations

Standard OT is expensive $\Rightarrow$ Approximate/regularized OT

- Entropy-regularized OT (Sinkhorn algorithm) [Cuturi, 2013]
- Sliced OT [Rabin et al., 2011]

The Curse of Dimensionality

In practice, we only observe finite samples $(x_i)_{i=1}^n \overset{i.i.d.}{\sim} \mu$, $(y_i)_{i=1}^n \overset{i.i.d.}{\sim} \nu$

$$\mathbf{W}_p(\mu_n, \nu_n) \quad \text{vs.} \quad \mathbf{W}_p(\mu, \nu)$$

How fast does $\mathbf{W}_p(\mu_n, \nu_n) \to \mathbf{W}_p(\mu, \nu)$ as $n \to +\infty$?

Sample complexity of $\mathbf{W}_p$ [Dudley, 1969]

If μ, ν are supported on a bounded domain of $\mathbb{R}^d$, $d > 2$,

$$\mathbb{E}\big[\,|\mathbf{W}_p(\mu_n, \nu_n) - \mathbf{W}_p(\mu, \nu)|\,\big] = \mathcal{O}(n^{-\frac{1}{d}})$$

with $\mathbb{E}$ taken over $(x_i, y_i)_{i=1}^n$.

Consequences:

- Higher dimension $d \Rightarrow$ slower convergence
- Accurate estimates may require prohibitively large n

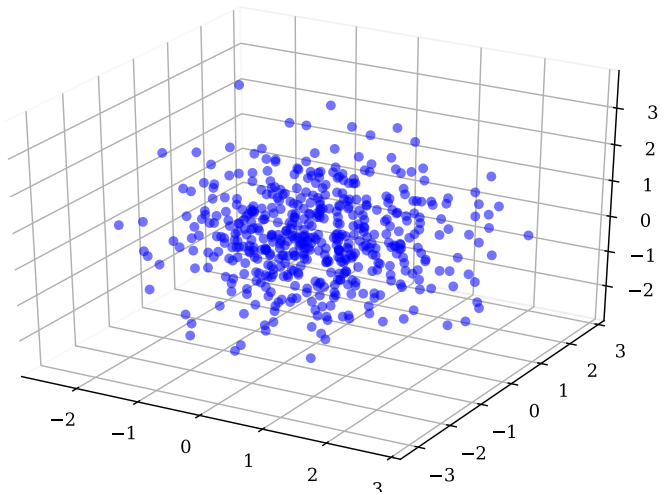
Sliced Optimal Transport

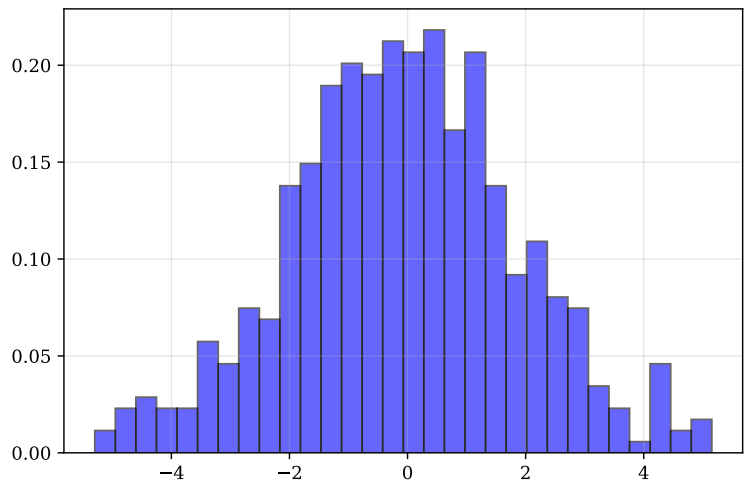
Slicing: From High to One Dimension

For $\theta = [\theta^{(1)}\, \theta^{(2)}\, \cdots\, \theta^{(d)}]^\top \in \mathbb{R}^d$ and $x = [x^{(1)}\, x^{(2)}\, \cdots\, x^{(d)}]^\top \in \mathbb{R}^d$, let

$$\langle \theta, x \rangle = \sum_{j=1}^{d} \theta^{(j)} x^{(j)} \in \mathbb{R}$$

project along $\theta \in \mathbb{R}^d$





$\mu = \frac{1}{n} \sum_{i=1}^{n} \delta_{x_i}$

$\mu^\theta = \frac{1}{n} \sum_{i=1}^{n} \delta_{\langle \theta, x_i \rangle}$

Sliced-Wasserstein Distance

The Sliced-Wasserstein Distance

Let $p \geq 1$. For two d-dimensional probability measures (μ, ν),

$$\mathbf{SW}_p(\mu, \nu) \;=\; \mathbb{E}_{\theta \sim \mathcal{U}(\mathbb{S}^{d-1})} \left[\mathbf{W}_p^p(\mu^\theta, \nu^\theta) \right]^{1/p}$$

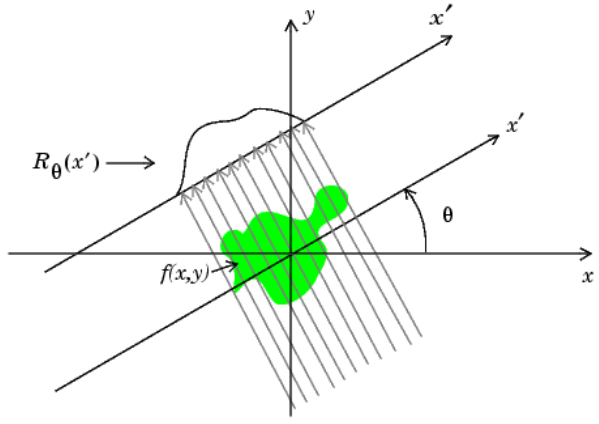
with,

- μ^θ, ν^θ : 1D projections of μ, ν along θ
- $\mathcal{U}(\mathbb{S}^{d-1})$: uniform distribution on the unit $(d-1)$-sphere,

$$\mathbb{S}^{d-1} = \{\theta \in \mathbb{R}^d \mid \|\theta\| = 1\}$$

Connection to the Radon Transform

$\mathbf{SW}_p$ can be expressed with the Radon transform of measures

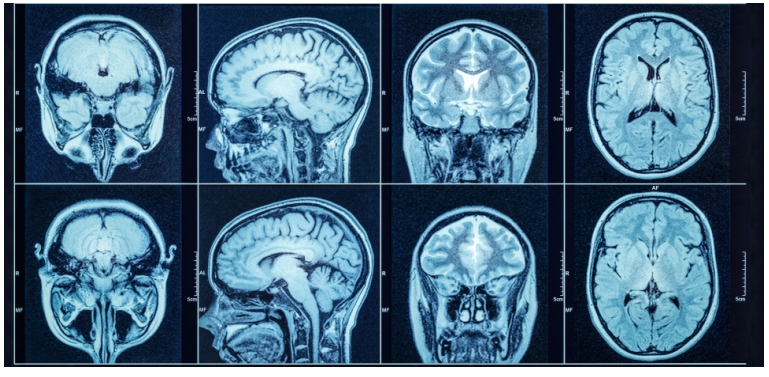


Radon transform (source)

Widely used for image reconstruction, *e.g.*, in tomography



CT scanner (source)



CT scans (source)

In Practice

$$\mathbf{SW}_p(\mu, \nu) \;=\; \left[\mathbb{E}_{\theta \sim \mathcal{U}(\mathbb{S}^{d-1})} \mathbf{W}_p^p(\mu^\theta, \nu^\theta)\right]^{1/p}$$

Recipe: Project high-dimensional distributions onto lines, sort (compute simple 1D distances), then aggregate

1. Sample $\theta \sim \mathcal{U}(\mathbb{S}^{d-1})$
2. Project to get μ^θ, ν^θ
3. Compute 1D Wasserstein distance $\mathbf{W}_p(\mu^\theta, \nu^\theta)$
4. Repeat k times (steps 1-2-3), then average

$$\widehat{\mathbf{SW}}_p(\mu, \nu) \;=\; \left(\frac{1}{k} \sum_{j=1}^{k} \mathbf{W}_p^p(\mu^{\theta_j}, \nu^{\theta_j})\right)^{1/p}$$

What Does Sliced OT Cost?

$$\widehat{\mathbf{SW}}_p(\mu,\nu) = \left(\frac{1}{k}\sum_{j=1}^{k}\mathbf{W}_p^p(\mu^{\theta_j},\nu^{\theta_j})\right)^{1/p}$$

If $\mu = \frac{1}{n}\sum_{i=1}^n x_i$, and $\nu = \frac{1}{n}\sum_{i=1}^n y_i$,

$$\mathbf{W}_p^p(\mu^{\theta_j},\nu^{\theta_j}) = \frac{1}{n}\sum_{i=1}^{n}\left|\langle\theta_j, x_{(i)}\rangle - \langle\theta_j, y_{(i)}\rangle\right|^p$$

with $x_{(1)} \leq \cdots \leq x_{(n)}$, $y_{(1)} \leq \cdots \leq y_{(n)}$

$\Rightarrow$ Computational complexity: $\mathcal{O}\big(\underbrace{kdn}_{\text{projecting}} + \underbrace{kn\log(n)}_{\text{sorting}}\big)$

Compared to standard OT

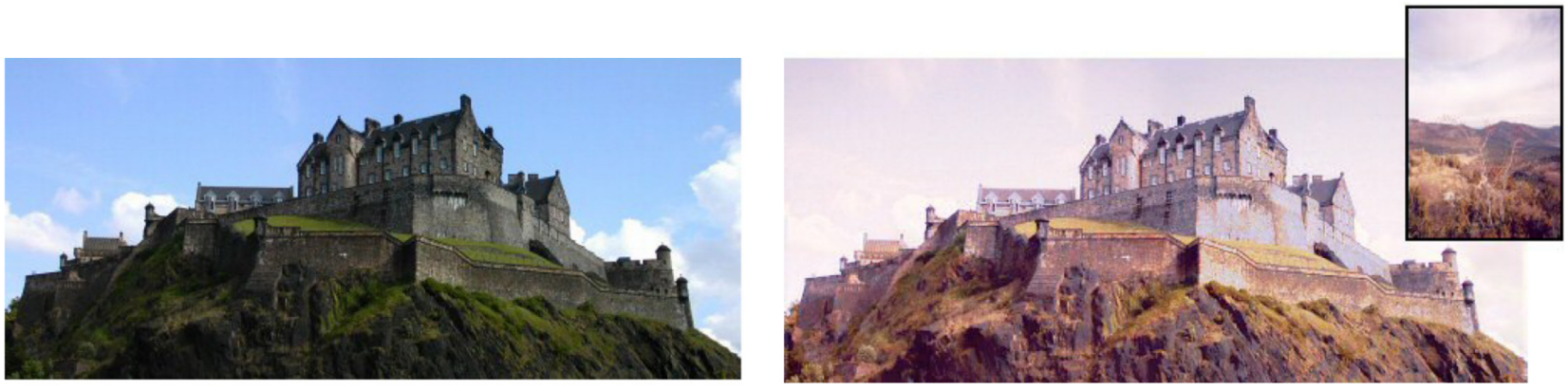
$\mathbf{W}_p(\mu,\nu)$	$\mathcal{O}(n^3)$
$\mathbf{SW}_p(\mu,\nu)$	$\mathcal{O}(kdn + kn\log n)$

$\Rightarrow$ SW is much cheaper for $n \gg d$ and moderate k

From Color Transfer...

Origins: Image processing

2006: Iterative Distribution Transfer (IDT): algorithm for color transfer, by iteratively matching 1D projections

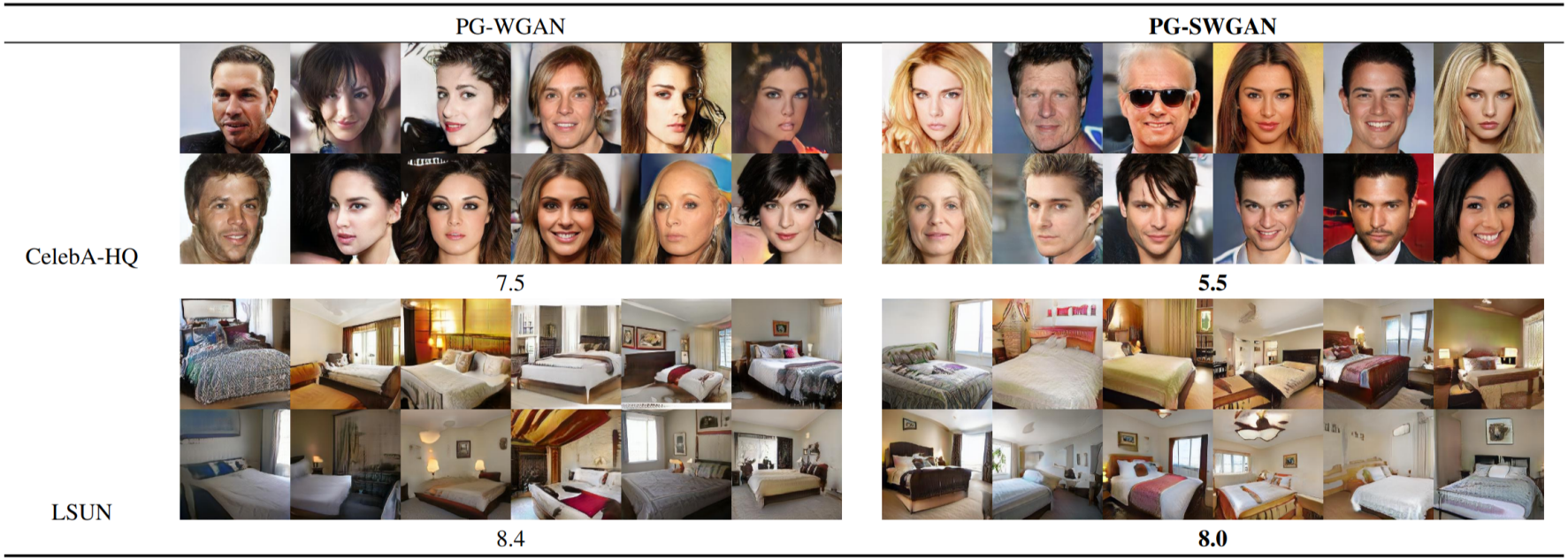


Color transfer done with IDT [Pitié et al., 2007]

2012: Introduction of Sliced-Wasserstein distance (Bernot et al.)

2013: Theoretical analysis and link to gradient flows (Bonnotte's PhD thesis)

... To Generative Modeling



Images generated by (Sliced-)Wass. GANs with FID scores [Wu et al., 2019]

2017: Wasserstein GANs / auto-encoders

2018: Sliced-Wasserstein GANs / auto-encoders [Deshpande et al., 2018; Wu et al., 2019; Kolouri et al., 2019b]

2018 – present: Growing theory and applications of SW in ML

Mathematical Properties of Sliced-Wasserstein Distance

Why Sliced-Wasserstein is a Distance

The Sliced-Wasserstein distance satisfies all metric axioms

Let $\mu, \nu, \xi \in \mathcal{P}(\mathbb{R}^d)$,

(Non-negativity) $\mathbf{SW}_p(\mu, \nu) \geq 0$

(Identity of indiscernibles) $\mathbf{SW}_p(\mu, \nu) = 0 \iff \mu = \nu$

(Symmetry) $\mathbf{SW}_p(\mu, \nu) = \mathbf{SW}_p(\nu, \mu)$

(Triangle inequality) $\mathbf{SW}_p(\mu, \nu) \leq \mathbf{SW}_p(\mu, \xi) + \mathbf{SW}_p(\xi, \nu)$

Comparison with the Wasserstein distance

Always smaller

$$\forall \mu, \nu \in \mathcal{P}(\mathbb{R}^d), \quad \mathbf{SW}_p^p(\mu, \nu) \leq c_{d,p} \mathbf{W}_p^p(\mu, \nu)$$

with $c_{d,p} = \frac{1}{d} \mathbb{E}_{\theta \sim \mathcal{U}(\mathbb{S}^{d-1})}[\|\theta\|_p^p] \leq 1$. If $p \geq 2$, $c_{d,p} \leq \frac{1}{d}$.

Equivalent on bounded supports

If $\mu, \nu \in \mathcal{B}(\mathbf{0}, R)$ (ball of radius R),

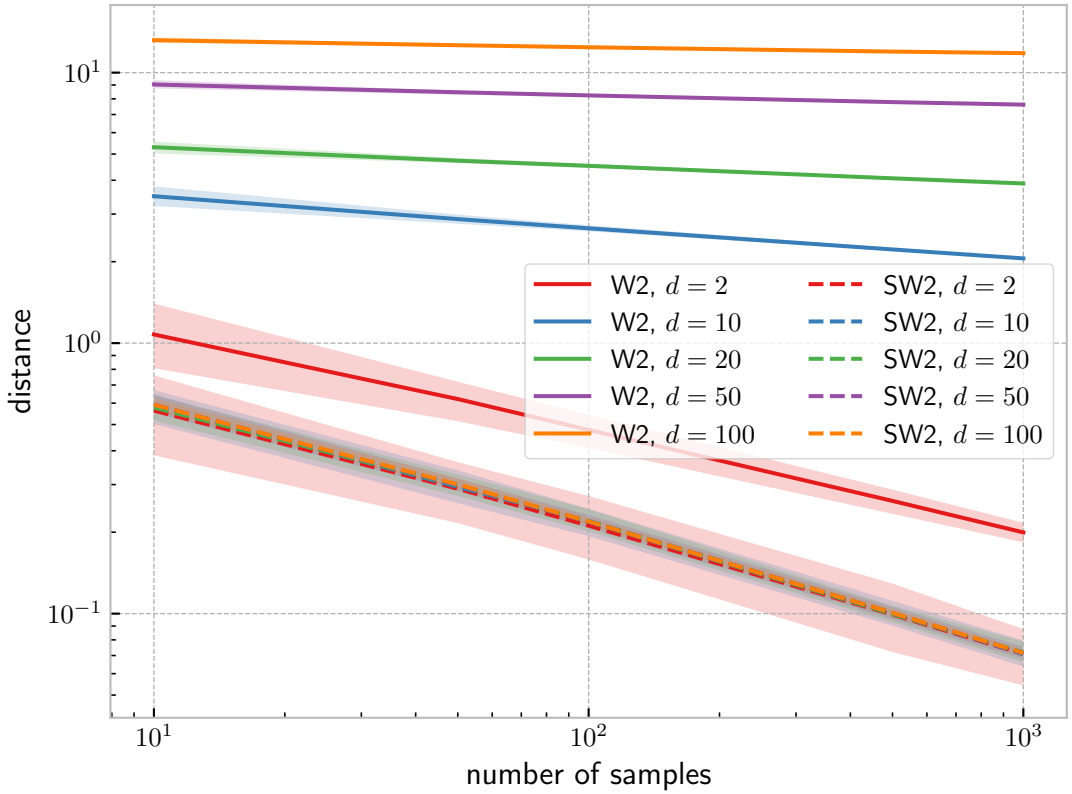
$$\mathbf{SW}_p^p(\mu, \nu) \leq c_{d,p} \mathbf{W}_p^p(\mu, \nu) \leq C_{d,p} R^{p-1/(d+1)} \mathbf{SW}_p(\mu, \nu)^{1/(d+1)}$$

Exact formula for Gaussians

If $\mu = \mathcal{N}(\mathbf{m}_1, \sigma_1^2 \mathbf{I}_d)$ and $\nu = \mathcal{N}(\mathbf{m}_2, \sigma_2^2 \mathbf{I}_d)$,

$$\mathbf{SW}_2(\mu, \nu) = \frac{1}{\sqrt{d}} \mathbf{W}_2(\mu, \nu)$$

Slicing Beats the Curse of Dimensionality?



$\mathbf{SW}_2$ vs. $\mathbf{W}_2$ between two sets of n i.i.d. samples from $\mathcal{N}(\mathbf{0}, \mathbf{I}_d)$

Sample complexity of $\mathbf{SW}_p$

For μ, ν supported on a bounded domain of $\mathbb{R}^d$,

$$\mathbb{E}\big[\,|\mathbf{SW}_p(\mu_n, \nu_n) - \mathbf{SW}_p(\mu, \nu)|\,\big] = \mathcal{O}(n^{-\frac{1}{2p}})$$

$\Rightarrow$ Dimension-free rate, unlike $\mathbf{W}_p$ $(\mathcal{O}(n^{-\frac{1}{d}}))$

Still Some Practical Challenges!

The previous rate applies to $\mathbf{SW}_p(\mu,\nu) = \left[\mathbb{E}_{\theta\sim\mathcal{U}(\mathbb{S}^{d-1})}\mathbf{W}_p^p(\mu^\theta,\nu^\theta)\right]^{1/p}$, which is intractable in practice

Practical approximation uses k random projections:

$$\widehat{\mathbf{SW}}_p^p(\mu,\nu) = \frac{1}{k}\sum_{j=1}^{k}\mathbf{W}_p^p(\mu^{\theta_j},\nu^{\theta_j})$$

Monte Carlo approximation error

$$\mathbb{E}\left[\,\left|\widehat{\mathbf{SW}}_p^p(\mu,\nu) - \mathbf{SW}_p^p(\mu,\nu)\right|\,\right] \leq \frac{1}{k}\,\mathrm{Var}_\theta\left[\mathbf{W}_p^p(\mu^\theta,\nu^\theta)\right]$$

Trade-off: more projections $\Rightarrow$ better accuracy, but higher computational cost

Beyond the Classical Sliced-Wasserstein Distance

Extensions of $\mathbf{SW}_p$

$$\mathbf{SW}_p(\mu, \nu) \;=\; \left[\mathbb{E}_{\theta \sim \mathcal{U}(\mathbb{S}^{d-1})}\, \mathbf{W}_p^p(\mu^\theta, \nu^\theta)\right]^{1/p}$$

- **Beyond uniform sampling**

Max SW [Deshpande et al., 2019; Paty and Cuturi, 2019], distributional and energy-based SW [Nguyen et al., 2021; Nguyen and Ho, 2023]

- **Beyond linear projections**
  - Generalized SW [Kolouri et al., 2019a]
  - On manifolds: Sphere [Bonet et al., 2023; Tran et al., 2024; Quellmalz et al., 2024], Cartan-Hadamard manifolds [Bonet et al., 2025]

- **Beyond Monte Carlo estimation**
  - Orthonormal sampling [Wu et al., 2019; Rowland et al., 2019]
  - Quasi-Monte Carlo [Nguyen et al., 2024; Sisouk et al., 2025]
  - Variance reduction via control variates [Leluc et al., 2024]
  - Concentration of measure approximation [Nadjahi et al., 2021]

For a broader overview: [Nguyen, 2025; Sisouk et al., 2025]

Beyond Uniform Sampling

$$\mathbf{SW}_p(\mu, \nu) = \left[\mathbb{E}_{\theta \sim \mathcal{U}(\mathbb{S}^{d-1})} \mathbf{W}_p^p(\mu^\theta, \nu^\theta)\right]^{1/p}$$

- **Beyond uniform sampling**

Max SW [Deshpande et al., 2019; Paty and Cuturi, 2019], distributional and energy-based SW [Nguyen et al., 2021; Nguyen and Ho, 2023]

- **Beyond linear projections**
  - Generalized SW [Kolouri et al., 2019a]
  - On manifolds: Sphere [Bonet et al., 2023; Tran et al., 2024; Quellmalz et al., 2024], Cartan-Hadamard manifolds [Bonet et al., 2025]
- **Beyond Monte Carlo estimation**
  - Orthonormal sampling [Wu et al., 2019; Rowland et al., 2019]
  - Quasi-Monte Carlo [Nguyen et al., 2024; Sisouk et al., 2025]
  - Variance reduction via control variates [Leluc et al., 2024]
  - Concentration of measure approximation [Nadjahi et al., 2021]

For a broader overview: [Nguyen, 2025; Sisouk et al., 2025]

Max Sliced Wasserstein distance

Let $p \geq 1$ and μ, ν be two probability measures on $\mathbb{R}^d$.

Max Sliced-Wasserstein distance [Deshpande et al., 2019]:

$$\text{max-}\mathbf{SW}_p(\mu, \nu) = \max_{\theta \in \mathbb{S}^{d-1}} \mathbf{W}_p(\mu^\theta, \nu^\theta)$$

Topological properties [Bayraktar and Guo, 2021]

- $\mathbf{SW}_p(\mu, \nu) \leq \text{max-}\mathbf{SW}_p(\mu, \nu) \leq \mathbf{W}_p(\mu, \nu)$
- $\lim_{k \to +\infty} \mathbf{W}_p(\mu^k, \mu) = 0 \Leftrightarrow \lim_{k \to +\infty} (\text{max-})\mathbf{SW}_p(\mu^k, \mu) = 0$, with $(\mu^k)_{k \in \mathbb{N}}$ a sequence of measures

How to compute max-$\mathbf{SW}_p(\mu, \nu)$?

- Projected gradient ascent [Kolouri et al., 2019a; Nietert et al., 2022]
- Riemannian optimizaion on $\mathbb{S}^{d-1}$ [Lin et al., 2020]

Beyond Linear Projections

$$\mathbf{SW}_p(\mu, \nu) = \left[\mathbb{E}_{\theta \sim \mathcal{U}(\mathbb{S}^{d-1})} \mathbf{W}_p^p(\mu^\theta, \nu^\theta)\right]^{1/p}$$

- **Beyond uniform sampling**

Max SW [Deshpande et al., 2019; Paty and Cuturi, 2019], distributional and energy-based SW [Nguyen et al., 2021; Nguyen and Ho, 2023]

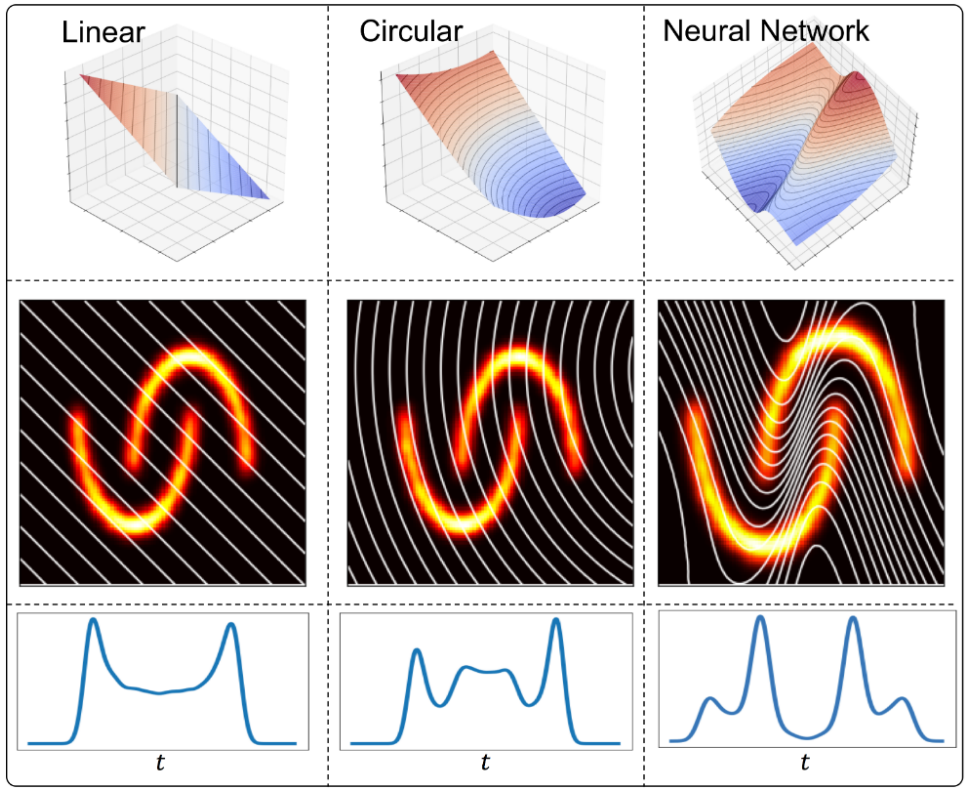
- **Beyond linear projections**
  - Generalized SW [Kolouri et al., 2019a]
  - On manifolds: Sphere [Bonet et al., 2023; Tran et al., 2024; Quellmalz et al., 2024], Cartan-Hadamard manifolds [Bonet et al., 2025]

- **Beyond Monte Carlo estimation**
  - Orthonormal sampling [Wu et al., 2019; Rowland et al., 2019]
  - Quasi-Monte Carlo [Nguyen et al., 2024; Sisouk et al., 2025]
  - Variance reduction via control variates [Leluc et al., 2024]
  - Concentration of measure approximation [Nadjahi et al., 2021]

For a broader overview: [Nguyen, 2025; Sisouk et al., 2025]

Generalized Sliced Wasserstein distances

Key idea: Replace linear projections $x \mapsto \langle \theta, x \rangle$ by more general functions $g^\theta : \mathbb{R}^d \to \mathbb{R}$ to define μ^θ, ν^θ



Examples of projection functions (Figure from Kolouri et al., 2019a)

Generalized Sliced-Wasserstein distances:

$$\mathbf{GSW}_p(\mu, \nu) \;=\; \left[\mathbb{E}_{\theta \sim \mathcal{U}(\mathbb{S}^{d-1})} \mathbf{W}_p(g^\theta_\sharp \mu, g^\theta_\sharp \nu)\right]^{1/p}$$

Beyond Monte Carlo Estimation

$$\mathbf{SW}_p(\mu, \nu) = \left[\mathbb{E}_{\theta \sim \mathcal{U}(\mathbb{S}^{d-1})} \mathbf{W}_p^p(\mu^\theta, \nu^\theta)\right]^{1/p}$$

- **Beyond uniform sampling**

Max SW [Deshpande et al., 2019; Paty and Cuturi, 2019], distributional and energy-based SW [Nguyen et al., 2021; Nguyen and Ho, 2023]

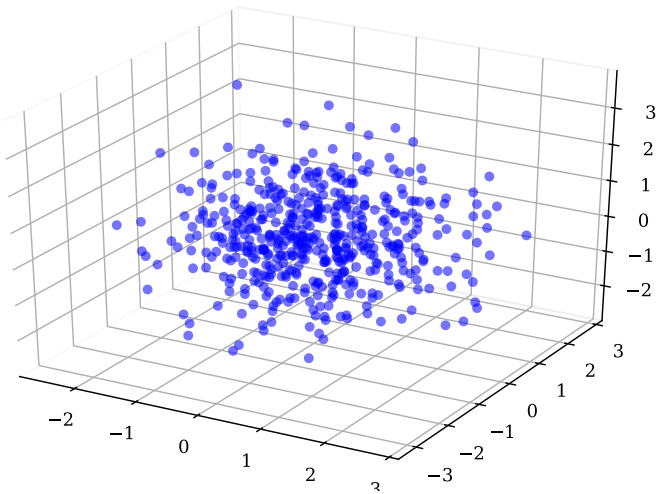
- **Beyond linear projections**
  - Generalized SW [Kolouri et al., 2019a]
  - On manifolds: Sphere [Bonet et al., 2023; Tran et al., 2024; Quellmalz et al., 2024], Cartan-Hadamard manifolds [Bonet et al., 2025]

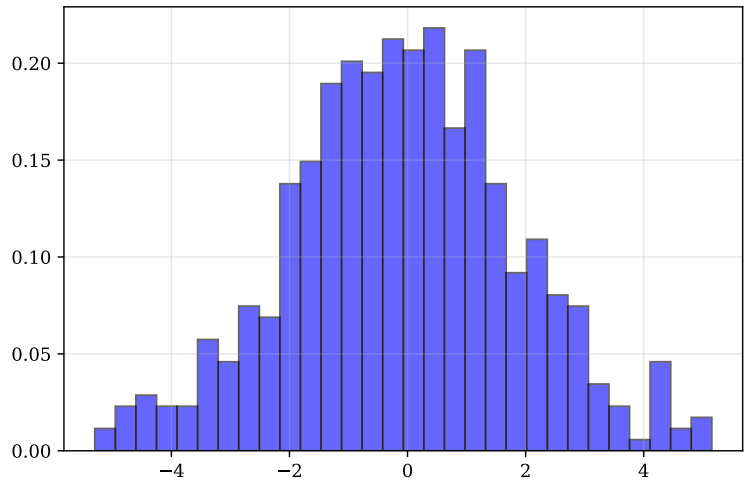
- **Beyond Monte Carlo estimation**
  - Orthonormal sampling [Wu et al., 2019; Rowland et al., 2019]
  - Quasi-Monte Carlo [Nguyen et al., 2024; Sisouk et al., 2025]
  - Variance reduction via control variates [Leluc et al., 2024]
  - Concentration of measure approximation [Nadjahi et al., 2021]

For a broader overview: [Nguyen, 2025; Sisouk et al., 2025]

Random Projections Lead to Gaussian Laws

project along $\theta \in \mathbb{R}^d$





$$\mu_d = \frac{1}{n}\sum_{i=1}^{n} \delta_{x_i} \qquad \mu_d^\theta = \frac{1}{n}\sum_{i=1}^{n} \delta_{\langle \theta, x_i \rangle}$$

Central limit theorem: Given $\theta \sim \mathcal{N}(\mathbf{0}, \Sigma)$, $(\mu_d^\theta)_{d\in\mathbb{N}^*}$ converges weakly to a Gaussian law in probability as $d \to +\infty$ (under some conditions on μ_d) [Sudakov, 1978; Diaconis and Freedman, 1984]

Gaussian Concentration Enables Fast $\mathbf{SW}_2$ Approximation

Concentration of 1D projections via $\mathbf{W}_2$ [Reeves, 2017]

Let $\mu_d \in \mathcal{P}_2(\mathbb{R}^d)$ and $\mathtt{m}_2(\mu_d) = \mathbb{E}_{X\sim\mu_d}\left[\|X\|^2\right]$. There is $C > 0$ s.t.,

$$\mathbb{E}_{\theta\sim\mathcal{N}(\mathbf{0},d^{-1}\mathbf{I}_d)}\left[\mathbf{W}_2^2\left(\mu_d^\theta, \mathcal{N}(0, d^{-1}\mathtt{m}_2(\mu_d))\right)\right] \leq C\,\Xi_d(\mu_d)$$

where $\Xi_d(\mu_d)$ depends on pairwise correlations of $d^{-1/2}X$, $X \sim \mu_d$.

Deterministic approximation of $\mathbf{SW}_2$ [Nadjahi et al., 2021]

$$\begin{aligned}&\left|\mathbf{SW}_2(\mu_d,\nu_d) - \mathbf{W}_2(\mathcal{N}(0, d^{-1}\mathtt{m}_2(\mu_d)), \mathcal{N}(0, d^{-1}\mathtt{m}_2(\nu_d)))\right| \\ &\leq C'\left\{\Xi_d(\mu_d) + \Xi_d(\nu_d)\right\}^{1/2}\end{aligned}$$

- If μ_d is centered with weak dependence, $\Xi_d(\mu_d) = \mathcal{O}(d^{-1/4})$
- Cost: $\mathcal{O}(dn)$ vs. $\mathcal{O}(kdn + kn\log n)$ for Monte Carlo

Transport Maps with Slicing

From Transport *Distances* to Transport *Maps*

- So far, we have focused on (sliced) OT distances
- Yet, in many applications, the useful object is the transport map

Back to the Monge problem

Find a transport map $T : \mathbb{R}^d \to \mathbb{R}^d$ solving

$$T = \arg \min_{S_{\#}\mu = \nu} \mathbb{E}_{x \sim \mu} \left[\|x - S(x)\|^2 \right].$$

Why transport maps matter:

- Domain adaptation/transfer learning [Courty et al., 2017]
- Computational biology (single-cell genomics) [Schiebinger et al., 2019; Bunne et al., 2024]
- Multivariate quantile and ranks to $\mathbb{R}^d$ [Chernozhukov et al., 2017], with recent applications to conformal prediction [Thurin et al., 2025]

Sliced-Based Transport Maps

$$\mathbf{SW}_p(\mu,\nu) \;=\; \left[\mathbb{E}_{\theta\sim\mathcal{U}(\mathbb{S}^{d-1})}\, \mathbf{W}_p^p(\mu^\theta,\nu^\theta)\right]^{1/p}$$

- Sliced OT defines $(T_\theta)_{\theta\in\mathbb{S}^{d-1}}$ s.t. $(T_\theta)_\sharp\mu^\theta = \nu^\theta$
- How to reconstruct a *global* $\widehat{T}:\mathbb{R}^d\to\mathbb{R}^d$ from $(T_\theta)_{\theta\in\mathbb{S}^{d-1}}$?

Slice-matching maps [Li and Moosmüller, 2024]

Let $P=[\theta_1|\theta_2|\cdots|\theta_d]\in\mathbb{R}^{d\times d}$ be an orthonormal basis of $\mathbb{R}^d$.

$$\forall x\in\mathbb{R}^d,\;\; T_P(x) = \sum_{i=1}^{d} T_{\theta_i}(\langle\theta_i,x\rangle)\theta_i$$

with T_{θ_i} : 1D transport map between μ^{θ_i} and ν^{θ_i}

Recent alternatives:

- Pivot-based SW plans [Mahey et al., 2023; Chapel et al., 2025]
- Expected SW plans ("*slice-and-lift*") [Liu et al., 2024]
- Theoretical analysis + numerical comparison [Tanguy et al., 2025]

References


Arjovsky, Martin, Soumith Chintala, and Léon Bottou (2017). "Wasserstein Generative Adversarial Networks". In: *Proceedings of the 34th International Conference on Machine Learning*. Ed. by Doina Precup and Yee Whye Teh. Vol. 70. Proceedings of Machine Learning Research. International Convention Centre, Sydney, Australia: PMLR, pp. 214–223.


Bayraktar, Erhan and Gaoyue Guo (2021). "Strong equivalence between metrics of Wasserstein type". In: *Electronic Communications in Probability* 26.none, pp. 1 –13. DOI: 10.1214/21-ECP383. URL: https://doi.org/10.1214/21-ECP383.


Bonet, Clément, Lucas Drumetz, and Nicolas Courty (2025). "Sliced-Wasserstein Distances and Flows on Cartan-Hadamard Manifolds". In: *Journal of Machine Learning Research* 26.32, pp. 1–76.


Bonet, Clément et al. (2023). "Spherical Sliced-Wasserstein". In: *The Eleventh International Conference on Learning Representations*.


Bunne, Charlotte et al. (2024). "Optimal transport for single-cell and spatial omics". In: *Nature Reviews Methods Primers* 4.1, p. 58. ISSN: 2662-8449. DOI: 10.1038/s43586-024-00334-2.


Chapel, Laetitia, Romain Tavenard, and Samuel Vaiter (2025). *Differentiable Generalized Sliced Wasserstein Plans*. arXiv: 2505.22049 [cs.LG].


Chernozhukov, Victor et al. (2017). "Monge–Kantorovich depth, quantiles, ranks and signs". In: *The Annals of Statistics* 45.1, pp. 223–256. DOI: 10.1214/16-AOS1450.


Courty, Nicolas et al. (2017). "Optimal Transport for Domain Adaptation". In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 39.9, pp. 1853–1865. DOI: 10.1109/TPAMI.2016.2615921.


Cuturi, Marco (2013). "Sinkhorn distances: Lightspeed computation of optimal transport". In: *Advances in Neural Information Processing Systems*, pp. 2292–2300.


Deshpande, Ishan, Ziyu Zhang, and Alexander G Schwing (2018). "Generative modeling using the sliced Wasserstein distance". In: *IEEE Conference on Computer Vision and Pattern Recognition*, pp. 3483–3491.


Deshpande, Ishan et al. (2019). "Max-Sliced Wasserstein Distance and its use for GANs". In: *IEEE Conference on Computer Vision and Pattern Recognition*.


Diaconis, Persi and David Freedman (1984). "Asymptotics of Graphical Projection Pursuit". In: *The Annals of Statistics* 12.3, pp. 793 –815. DOI: 10.1214/aos/1176346703.


Dudley, Richard M. (Feb. 1969). "The speed of mean Glivenko-Cantelli convergence". In: *Ann. Math. Statist.* 40.1, pp. 40–50. DOI: 10.1214/aoms/1177697802.


Kolouri, Soheil et al. (2019a). "Generalized Sliced Wasserstein Distances". In: *NeurIPS 2019*. Ed. by H. Wallach et al. Vol. 32. Curran Associates, Inc.


Kolouri, Soheil et al. (2019b). "Sliced Wasserstein Auto-Encoders". In: *International Conference on Learning Representations*.


Leluc, Rémi et al. (2024). “Sliced-Wasserstein Estimation with Spherical Harmonics as Control Variates”. In: *Proceedings of the 41st International Conference on Machine Learning*. Ed. by Ruslan Salakhutdinov et al. Vol. 235. Proceedings of Machine Learning Research. PMLR, pp. 27191–27214.


Li, Shiying and Caroline Moosmüller (2024). “Approximation properties of slice-matching operators”. In: *Sampling Theory, Signal Processing, and Data Analysis* 22.1, p. 15. ISSN: 2730-5724. DOI: 10.1007/s43670-024-00089-7.


Lin, Tianyi et al. (2020). “Projection robust Wasserstein distance and Riemannian optimization”. In: *Proceedings of the 34th International Conference on Neural Information Processing Systems*. NeurIPS 2020. Vancouver, BC, Canada: Curran Associates Inc. ISBN: 9781713829546.


Liu, Xinran et al. (2024). *Expected Sliced Transport Plans*. arXiv: 2410.12176 [cs.LG].


Mahey, Guillaume et al. (2023). “Fast Optimal Transport through Sliced Generalized Wasserstein Geodesics”. In: *Thirty-seventh Conference on Neural Information Processing Systems*.

 Monge, Gaspard (1781). “Mémoire sur la théorie des déblais et des remblais”. In: *Mem. Math. Phys. Acad. Royale Sci.*, pp. 666–704.

 Nadjahi, Kimia et al. (2021). “Fast Approximation of the Sliced-Wasserstein Distance Using Concentration of Random Projections”. In: *NeurIPS 2021*.

 Nguyen, Khai (2025). *An Introduction to Sliced Optimal Transport*. arXiv: 2508.12519 [stat.ML].

 Nguyen, Khai, Nicola Bariletto, and Nhat Ho (2024). “Quasi-Monte Carlo for 3D Sliced Wasserstein”. In: *The Twelfth International Conference on Learning Representations*.

 Nguyen, Khai and Nhat Ho (2023). “Energy-Based Sliced Wasserstein Distance”. In: *Thirty-seventh Conference on Neural Information Processing Systems*.

 Nguyen, Khai et al. (2021). “Distributional Sliced-Wasserstein and Applications to Generative Modeling”. In: *International Conference on Learning Representations*.


Nietert, Sloan et al. (2022). “Statistical, Robustness, and Computational Guarantees for Sliced Wasserstein Distances”. In: *Advances in Neural Information Processing Systems*. Ed. by Alice H. Oh et al.


Paty, François-Pierre and Marco Cuturi (2019). “Subspace Robust Wasserstein distances”. In: *International Conference on Machine Learning*.


Pitié, François, Anil C. Kokaram, and Rozenn Dahyot (2007). “Automated colour grading using colour distribution transfer”. In: *Computer Vision and Image Understanding* 107.1. Special issue on color image processing, pp. 123–137. ISSN: 1077-3142. DOI: https://doi.org/10.1016/j.cviu.2006.11.011.


Quellmalz, Michael, Léo Buecher, and Gabriele Steidl (2024). “Parallelly Sliced Optimal Transport on Spheres and on the Rotation Group”. In: *Journal of Mathematical Imaging and Vision* 66.6, pp. 951–976. ISSN: 1573-7683. DOI: 10.1007/s10851-024-01206-w.

Rabin, Julien et al. (2011). “Wasserstein Barycenter and Its Application to Texture Mixing”. In: *Scale Space and Variational Methods in Computer Vision*. Ed. by Alfred M. Bruckstein et al. Berlin, Heidelberg: Springer Berlin Heidelberg, pp. 435–446. ISBN: 978-3-642-24785-9.

Reeves, Galen (2017). “Conditional central limit theorems for Gaussian projections”. In: *2017 IEEE International Symposium on Information Theory (ISIT)*, pp. 3045–3049. DOI: 10.1109/ISIT.2017.8007089.

Rowland, Mark et al. (2019). “Orthogonal Estimation of Wasserstein Distances”. In: *Proceedings of the Twenty-Second International Conference on Artificial Intelligence and Statistics*. Ed. by Kamalika Chaudhuri and Masashi Sugiyama. Vol. 89. Proceedings of Machine Learning Research. PMLR, pp. 186–195.

Schiebinger, Geoffrey et al. (2019). “Optimal-Transport Analysis of Single-Cell Gene Expression Identifies Developmental Trajectories in Reprogramming”. In: *Cell* 176.4, 928–943.e22. ISSN: 0092-8674. DOI: https://doi.org/10.1016/j.cell.2019.01.006.

 Sisouk, Keanu, Julie Delon, and Julien Tierny (2025). "A User's Guide to Sampling Strategies for Sliced Optimal Transport". In: *Transactions on Machine Learning Research*. Survey Certification. ISSN: 2835-8856.

 Sudakov, Vladimir Nikolaevich (1978). "Typical distributions of linear functionals in finite dimensional spaces of high dimension". In: *Soviet Math. Dokl.* 19.6, pp. 1578 –1582.

 Tanguy, Eloi, Laetitia Chapel, and Julie Delon (2025). *Sliced Optimal Transport Plans*. arXiv: 2508.01243 [math.OC].

 Thurin, Gauthier, Kimia Nadjahi, and Claire Boyer (2025). "Optimal transport-based conformal prediction". In: *Forty-second International Conference on Machine Learning*.

 Tolstikhin, I. et al. (May 2018). "Wasserstein Auto-Encoders". In: *6th International Conference on Learning Representations (ICLR)*.

 Tran, Huy et al. (2024). "Stereographic Spherical Sliced Wasserstein Distances". In: *ICML*.

 Wu, Jiqing et al. (2019). "Sliced Wasserstein Generative Models". In: *IEEE Conference on Computer Vision and Pattern Recognition*.